

LLM Data Forge: Distributed Data Curation and Quality Optimization for Efficient LLM Pretraining

CSCI 6996 Independent Study

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Abstract:

Massive training datasets are required for big language models, but these datasets frequently include inconsistent data sources, redundant text, duplicate content, low-quality data, and boilerplate. Storage, preprocessing, and computing expenses are increased when processing and training on redundant or poor-quality data. LLM Data Forge proposes a distributed Big Data pipeline for collecting, processing, filtering, deduplicating, and constructing LLM training datasets. The project will use distributed processing techniques such as Apache Spark, partitioning, Parquet, hashing, Min Hash, and Locality-Sensitive Hashing to create multiple versions of training datasets.

A controlled small/medium-scale language model will be used to assess the resultant datasets in order to investigate the effects of data composition, deduplication, and quality on training efficiency and model performance.

Keywords:

Big Data, Large Language Models, Distributed ETL, Apache Spark, Data Curation, Data Quality, Deduplication, Min Hash, Locality-Sensitive Hashing, Dataset Composition, Tokenization, LLM Pretraining, Data Efficiency, Compute Efficiency

Problem Statement:

Large Language Models require large volumes of training data, but these datasets often contain duplicate, repetitive, noisy, or low-quality content. Processing unnecessary data increases storage, processing time, and computational cost. This project addresses the challenge of identifying and improving the quality of LLM training data using scalable distributed data-processing techniques.

Objective:

The objective is to develop a distributed data-curation pipeline that filters, deduplicates, and organizes large-scale LLM training data, and evaluates the impact of these techniques on training efficiency and model performance.

End Goal:

The end goal is to determine whether better-curated training data can reduce data and computational requirements while maintaining or improving LLM performance, providing practical insights into more efficient LLM training.

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